Al-Balqa Applied University
Faculty of Engineering Technology
Department of Electrical Engineering

Signals and Systems
Midterm Exam
2nd Semester 2024-2025

Student Name: .....

Student ID: .....

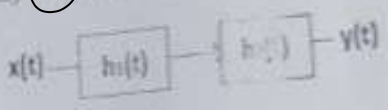
Q1. Select the correct answer for the following questions:

1. The result of multiplying $x(t) = t^2$ and $g(t) = \sin(3t)$ is..... function.
a. Odd b. Even c. Neither even nor odd. d. Can not be determined

2. The convolution between $x(t) = e^{-t}u(t-4)$ and $\delta(t+4)$ for $t > 1$ is given as.....
a. $e^{-4t}u(-4)$ b. e^{-4} c. $e^{-4t}u(t-8)$ d. $\frac{1}{e^{4(t+1)}}$ e. $e^{4t}u(t-8)$ f. $e^{4t-8}u(t)$

3. For LTI system, a ..... system is a ..... system.
a. Memoryless, causal. b. Time-invertible, causal. c. Stable, memoryless. d. Causal, stable. e. Linear, stable.

4. The overall impulse response $h(t)$ of the following cascade LTI system, if $h_1(t) = \delta(t-X)$ and $h_2(t) = \delta(t-5)$, is given as...
a. $h(t) = \delta(2)$ b. $h(t) = \delta(t+2)$ c. $h(t) = \delta(-2)$ d. $h(t) = \delta(-8)$ e. $h(t) = \delta(t-8)$



5. One of the following signals is considered as causal system, which is .....
a. $y[n] = 2x[n] - 3x[n+1]$
b. $y[n] = 2x[n-7] - 3x[n+1]$
c. $y[n] = 2x[n] - 3x[n+1]$
d. $y[n] = 2x[n] + 3$
e. None of them

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6. The result of $\cos(2t) + \cos(-2t)$ is.....
a. $2\cos(2t)$ b. $0.5\cos(t)$ c. $2\cos(t)$ c. $2\cos(4t)$ d. $0.5\cos(2t)$ e. None of them

7. The following signal $x[n] = \sin(\frac{2\pi n}{5})$ is considered as .....
a. Periodic, discrete, and $N = 5/2$
b. Periodic, discrete, and $N = 5$
c. Aperiodic, continuous, and $N = 5/2$
d. Aperiodic, discrete, and $N = 2/5$
e. Aperiodic, discrete and $N = 5$

8. The system ..... is considered as a memory system.
a. $y(n) = x(n) + x(n+1)$ b. $y(t) = x^2(t) + x(t)$ c. $y(t) = x(t)x(t-1)$ d. a and b e. a and c

9. Which of the following is NOT true?
a. $t^2\delta(t) = 0$ b. $\cos(t)\delta(t) = \delta(t)$ c. $\sin(t)\delta(t-\pi) = \delta(t-\pi)$ d. $\int_{-\infty}^{\infty}(2t^2+1)\delta(t)dt = 1$

10. The result of $\int_0^4 3\delta(t+1)\,dt$ is given as.....
a. 3 b. 3t c. 12t d. 0 e. 12

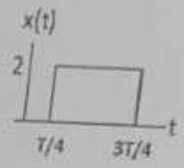
11. The power of the signal $s(t) = 6\sin\left(5\pi t - \frac{\pi}{4}\right) + 4\cos(17\pi t)$ is.....
a. 10 b. 100 c. 52 d. 26 e. None of them

12. The fundamental period of the signal $x(t) = 2\sin\left(\frac{8t}{2}\right) + \cos(4t/6 + 40°)$ is.....

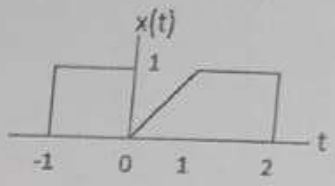
a. 4π b. 24π c. $\pi/4$ d. $6\pi/8$ e. $4\pi/2$ f. None of them

13. The signal $x(t)$ that represents the following figure is.....

a. $-2\,u\left(t-\frac{T}{2}\right) + 4\,u(t-T/2)$ b. $2\,rect\left(\frac{t-\frac{T}{4}}{T}\right)$ c. $2\,rect\left(\frac{t-\frac{T}{2}}{T/2}\right)$ d. $2\,rect\left(\frac{t}{T}-\frac{1}{4}\right)$



Q2. Draw the even component of the signal $x(t)$ shown below. (Show all necessary steps)

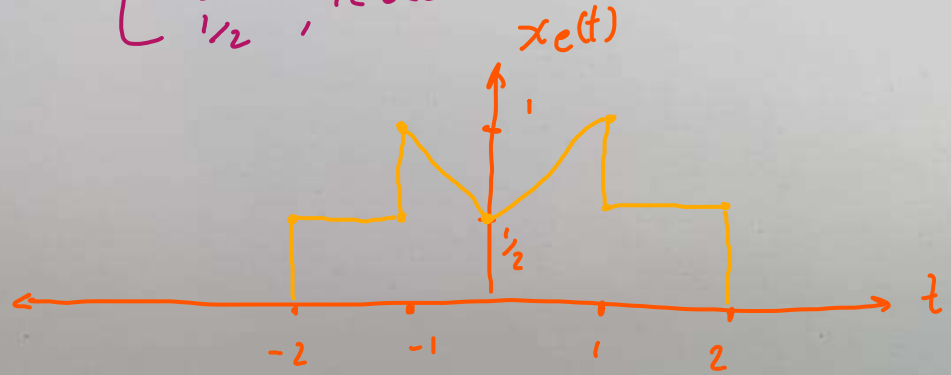


$$x(t) = \begin{cases} 1, & -1 < t < 0 \\ t, & 0 < t < 1 \\ 1, & 1 < t < 2 \\ 0, & o.w \end{cases}$$

$$x(-t) = \begin{cases} 1, & -1 < -t < 0 \\ -t, & 0 < -t < 1 \\ 1, & 1 < -t < 2 \\ 0, & o.w \end{cases} \Rightarrow x(+t) = \begin{cases} 1, & -2 < t < -1 \\ -t, & -1 < t < 0 \\ 1, & 0 < t < 1 \\ 0, & o.w \end{cases}$$

$$x_e(t) = \frac{x(t) + x(-t)}{2} = \begin{cases} 1/2, & -2 < t < -1 \\ \frac{1-t}{2}, & -1 < t < 0 \\ \frac{1+t}{2}, & 0 < t < 1 \\ 1/2, & 1 < t < 2 \end{cases}$$

$(-2, 1/2)$, $(-1, 1/2)$
$(-1, 1)$, $(0, 1/2)$
$(0, 1/2)$, $(1, 1)$
$(1, 1/2)$, $(2, 1/2)$



Q1 Answers:

1	2	3	4	5	6	7	8	9	10	11	12	13

Good Luck!

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1) $x(t) = t^2 \longrightarrow$ even $\Big]$ $\longrightarrow$ odd $\quad$ A.

$g(t) = \sin 3t \longrightarrow$ odd $\Big]$ x

2) $x(t) * \delta(t+4) = x(t+4) = y(t)$

$y(t) = e^{-(t+4)}\, u(t+4-4) = e^{-(t+4)}$ for $t > 0$.

$y(t) = e^{-(t+4)}$ $\quad u(t-1) \longrightarrow y(t) = \dfrac{1}{e^{t+4}}$. $\quad$ D.

for $t > 1$

3) LTI system : A.

A. Memoryless $\xrightarrow{\text{then}}$ Causal

(instantaneous) (indep of future). ✓

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B. Time Invertiable $\xrightarrow{\text{then}}$ Causal ✗

$y(t) = x(-t)$ $\quad y(-1) = x(1)$

C. Stable $\xrightarrow{\text{then}}$ memoryless

$h(t) = e^{t} u(t)$ $\quad y(t) = x(t) * h(t)$ ✗

$|y(t)| \le M |h(t)|$ $\quad = \int_{-\infty}^{\infty} x(\tau)\, e^{t-\tau}\, u(t-\tau)\, d\tau$

BIBO. $\quad$ has memory ← $t > \tau$

⋮

4) $h(t) = h_1(t) * h_2(t)$

$= \int_{-\infty}^{\infty} \delta(\tau - 3)\ \delta(t-5-\tau)\, d\tau$ E.

$= \delta(t-5-3) = \delta(t-8)$.

5) causal system :-

D. $y[0] = 2x[0] + 3$

6) $\cos 2t + \cos(-2t) = \cos 2t + \cos 2t = 2\cos 2t$. A.

7) $x[n] = \sin\left(\frac{2\pi}{5} t\right)$ B.

discrete, periodic, $N = \frac{2\pi}{\omega_0} = \frac{2\pi}{1} \times \frac{5}{2\pi} = 5$.

8) memory system :- E. A & C.

A. $y(0) = x(0) + x(1)$, C. $y(0) = x(0)\, x(-1)$.

9) NOT True: $\sin t\ \delta(t-\pi) = \sin\pi\ \delta(t-\pi)$

$= -\delta(t-\pi) \neq \delta(t-\pi)$ C.

10) $\int_0^4 3\delta(t-(-1))\, dt = 0 \longrightarrow t = -1 \notin [0,4]$. D.

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11) $P\{A\sin(\omega_0 t+\theta)\} = A^2/2$

$P\{A\cos(\omega_0 t+\theta)\} = A^2/2$

$$P_{tot} = \frac{6^2}{2} + \frac{4^2}{2} = 18+8 = 26 . \quad D.$$

12) $x(t) = 2\sin(4t) + \cos\left(\frac{2}{3}t + 40^\circ\right)$

$$T_1 = \frac{2\pi}{4} = \frac{\pi}{2}, \quad T_2 = \frac{2\pi}{1} \times \frac{3}{2} = 3\pi$$

$$\frac{T_1}{T_2} = \frac{\pi}{2} \times \frac{1}{3\pi} = \frac{1}{6} \longrightarrow \quad T_2 = T_0$$

$T_0 = 3\pi$, therefore f. none of the above.

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13) All answers are wrong.

$$x(t) = 2\,\mathrm{rect}\left(\frac{t - T/2}{T/2}\right).$$

closest answer is C.